COLX 531 & 585

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Motivation

Who is responsible for what goes into a model?

- Modern LLMs and VLMs are trained on massive corpora that may include clinical records, personal images, and copyrighted text often **without explicit consent.**
- Data owners and regulators need a way to **verify** whether a specific example influenced a model's training.

Membership Inference Attacks (MIAs) are the canonical privacy-auditing tool: given only query access to a model, decide whether a data point was in its training set.

Understanding MIA also reveals **how much models memorize** — a fundamental question in machine learning safety.

Definition

Given a sample x and query access to a trained model M, predict whether x was in M's training set.

Output: membership score

- Score near 1 = member
- Score near 0 = non-member

Two tasks in this work

- Task 1: LLM finetuned on medical notes
- Task 2: VLM finetuned on image-caption pairs

Both are evaluated in a **gray-box setting**: access to model outputs such as logits and loss, but not internal weights.

Previous Works - What has been tried — and where the gaps are.

Do Membership Inference Attacks Work on Large Language Models? - Duan et al. (2024)

Showed that popular MIA methods barely outperform random guessing on pretrained LLMs because large-scale training prevents strong memorization. This motivates our focus on the finetuned regime.

Membership Inference Attacks against Language Models via Neighbourhood Comparison - Mattern et al. (2023)

Proposed the Neighbourhood Attack: compare a sample's loss to perturbed neighbors to detect memorization without a reference dataset. We extend this to both text and image modalities in our VLM attack.

M^4I: : Multi-modal Models Membership Inference - Hu et al. (2022)

Introduced the first MIA framework for multimodal models, using ROUGE/BLEU similarity and a custom feature extractor. We modernize it by replacing the extractor with off-the-shelf CLIP and combining it with neighborhood signals.

Data

Task 1

Split	N	Mean Length	Max	Min
Train	50,000	2016.88	5650	752
Validation	10,000	2010.69	4886	836
Test	15,000	2021.14	5382	787

Task 2

Split	N	Mean	Med.	Max	Min	SD
Train	6,000	261.93	246	813	98	89.24
Validation	1,200	258.05	241	737	104	86.61
Test	6,000	247.19	232	1079	82	91.35

Methods - MIA on Finetuned Language Models

Method 1: Base Model

Assigns scores using raw cross-entropy loss.

Key: Members have lower loss.

Method 2: Min-K % Prob

Averages log-probs of **K%** most surprising tokens.

Key: Members have higher prob.

Method 3: Casing Attack

Measures loss increase from lowercasing text.

Key: Members suffer larger loss spikes from perturbation.

Method 4: Reference Model

Subtracts fine-tuned loss from base loss.

Key: Members show a larger loss drop.

Method 5: Neighborhood Attack

Extracts 16 cross-model features and token-drop neighbors.

Key: Characteristic loss gap between models pairs.

Experiments & Results (LLM)

Milestone	**Method**	**AUC**	**TPR@FPR=0.1**
M1	Raw Loss (Baseline)	0.5839	0.1552
M2	Reference Model (Loss Ratio)	0.6701	0.2214
M2	Min-K% Prob (K=20%)	0.6166	0.1682
M2	Casing Attack	0.5887	0.1566
M3	**Neighborhood Attack (GBC)**	**0.9071**	**0.6966**

Methods - MIA on Visual Language Model

Method 1: Raw Loss Baseline

Assigns scores using cross-entropy loss on assistant tokens.

Key: Members have lower loss.

Method 2: Neighborhood Attack

Generates perturbed text/image neighbors and measures loss gaps across finetuned and base models.

Key: Members sit at sharper local minima.

Method 3: Multi-Feature Min-K% Prob

Extracts 7 token-level features (loss, perplexity, Min-K%, zlib ratio, etc.) with Logistic Regression.

Key: Members have higher prob on hardest tokens.

Method 4: Lightweight Complexity Calibration

Same 7 features as Method 3 but with XGBoost to capture non-linear interactions.

Key: Non-linear classifier exploits feature interactions.

Methods - MIA on Visual Language Model

Method 5: M^4I-CLIP Attack

Generates text via SmolVLM, compares to ground truth using ROUGE and CLIP cosine similarities.

Key: Members have tighter image-text alignment in CLIP space.

Method 6: Hybrid MIA + LiRA

Supersets Method 5 with LiRA cross-model features and generation consistency signals (33 features).

Key: Combines data-level, model-level, and output-level signals.

Method 7: Neighborhood + MIA Combined

Combines loss ratio (finetuned vs base) with text/image perturbation gaps in 20 features.

Key: Loss ratio + perturbation sensitivity are complementary.

Experiments & Results (VLM)

Milestone	**Method**	**AUC**	**TPR@FPR=0.1**
M4	Raw Loss (Baseline)	0.5008	0.1200
M5	Multi-Feature Min-K% Prob	0.5127	0.0800
M5	Lightweight Complexity Calibration	0.5605	0.1800
M5	Neighborhood Attack	0.5988	0.1867
M6	M4I-CLIP Attack	0.8093	0.3000
M6	Hybrid MIA + LiRA	0.8061	0.2633
M6	**Neighborhood + MIA Combined**	**0.8486**	**0.3884**

Conclusion and Future Work

Multiple Model Success

Combining multiple individually weak membership signals.

Lifted validation AUC to 0.91 (LLM) and 0.85 (VLM), proving single-statistic attacks are insufficient for instruction-tuned models.

Modality-Specific Leaks

Effective membership signals are dependent on modality

Cross-model loss gaps dominate for text models; cross-modal alignment (CLIP) is the primary leak for vision-language models.

Efficiency & Scalability

Developing "inference-light" frameworks to reduce reliance on resource-intensive extraction processes.

Replacing heavy embedding models will bypass computationally and time intensive inferences.

Advanced Probing & Attribution

Moving beyond simple word-drop to more sophisticated semantic modifications and modality-specific analysis.

Identify if leaks occur primarily through images or text and test if these signals generalize various domains.

Thank You for Watching